

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Metallurgical & Materials Engineering

2. Subject Name: Machine learning applications in materials engineering laboratory

L-T-P: 0-0-3

Credits: 2

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

(1) Overview of machine learning tools: Introduction to numpy and scipy; basic operations on arrays and matrices; creating and manipulating data frames in pandas

(2) Data preparation and cleaning: Reading various file formats in python; imputation methods; standardization and normalization; quantile transformation

(3) Feature extraction and dimensionality reduction: Plotting correlation heatmaps; Principle component analysis in scikit-learn

(4) Linear regression and classification models: Logistic regression and SVM.

(5) General learning of discriminative models: Ridge regression and LASSO in scikit-learn; Methods of hyperparameter tuning; Cross-validation

(6) Neural networks: Creating multilayer ANN with scikit-learn; demonstration of PyTorch

(7) Ensemble learning: Realization of bagging in scikit-learn; Implementation of Random forest

(8) Data mining in materials engineering: Introduction and access to various materials databases like COD, ICSD, Pauling file, Materials project, etc.; Overview of Pymatgen.

Note: Datasets should involve systems like alloys, semiconductors, ceramics, polymers, etc., to showcase the wide applicability of the methods.

Reference Books:

1) Lee Vaughan, *Python Tools for Scientists: An Introduction to Using Anaconda, Jupyterlab, and Python's Scientific Libraries*, No Starch Press.

2) S. Raschka, Y. Liu, and V. Mirjalili, *Machine Learning with PyTorch and Scikit-Learn*, Packt Publishing.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of hours
1.	Overview of machine learning tools	4
2.	Data preparation and cleaning	4
3.	Feature extraction and dimensionality reduction	3
4.	Linear regression and classification models	4
5.	General learning of discriminative models	6
6.	Neural networks	6
7.	Ensemble learning	6
8.	Data mining in materials engineering	3
Total number of hours		36